

# Personalized Adaptive Mental-Health Detection (PAMHeD):

An experiment on leveraging behavioral baselines for better signal detection

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2026

## Abstract

**Problem Statement:** Current social media mental health detection models rely heavily on static textual analysis, suffering from poor cross platform generalizability and an inability to distinguish between genuine pathology and platform specific artifacts.

**Solution Method:** We introduce PAMHeD, a personalized approach that establishes a behavioral baseline for each user (posting rhythm, sentiment volatility, linguistic style) and trains models on the deviation (delta) from that baseline rather than raw content alone. The architecture fuses mental-roberta-base embeddings with engineered behavioral features, optimized via a One vs Rest (OvR) binary classification strategy to isolate specific conditions.

**Results Statement:** Initial 4 class experiments revealed that behavioral features effectively capture a “signal of normalcy,” significantly improving Control detection. In targeted binary tasks, the personalized “Screener” model outperformed the text only baseline (F1 0.69 vs 0.67). However, cross platform validation on the eRisk Reddit dataset revealed a critical trade off: text only linguistic markers generalize broadly (higher recall), while personalized behavioral signals act as a highly precise (67% precision), conservative filter that successfully captures universal biological symptoms (e.g., insomnia, emotional dysregulation) despite differing platform mechanics.

**Keywords:** Computational Psychiatry, NLP, Behavioral Modeling, Personalization, SHAP Explainability, Cross Platform Generalization

## 1. Introduction

Mental health screening at the population level is frequently constrained by limited clinical resources and stigma. Consequently, researchers have increasingly turned to public social media streams as a real time window into user psychological status. However, current computational approaches predominantly rely on “one size fits all” static models that fail to account for individual personality differences and baseline behaviors. The problem with such static analysis is that it misses crucial contextual nuances; a historically quiet user suddenly posting frequently signals a fundamentally different psychological shift than an already active user maintaining their volume. Furthermore, the reliance on heuristic labeling and the overfitting to platform specific artifacts, such as Twitter specific slang versus Reddit’s long form discourse, severely hinder the robustness and real world applicability of these models, often resulting in systems that learn spurious correlations rather than true clinical markers.

In this study, we hypothesize that a user’s deviation from their own established behavioral baseline is a significantly more robust predictor of mental health status than the semantic content of their posts alone. To investigate this, we constructed a system to com-

pare a text only Baseline model against a Personalized (Delta) model, evaluating them across both multi class and One vs Rest (OvR) binary settings, and subsequently testing their robustness on an out of domain dataset. Initially, we expected the Delta model to uniformly outperform the Baseline across all pathological classes in a multi class setting. However, actual results demonstrated that while the Delta model excelled at identifying healthy controls by leveraging stability markers, it struggled with inter class confusion among pathologies due to overlapping signals of instability. This finding necessitated a pivot to an OvR binary approach, which successfully yielded highly sensitive “Expert” models. Ultimately, cross platform testing revealed an unexpected but profound insight: while linguistic features are highly universal across platforms, personalized behavioral features are mechanically platform dependent but highly precise when capturing underlying biological markers, such as circadian disruption.

## 2. Background & Related Work

Over the past decade, researchers have increasingly looked to public social media streams as a rich, real

time lens on population mental health. The pivotal CLPsych 2015 shared task (Coppersmith et al., 2015) first demonstrated that simple lexical and n gram features applied to users who self disclose a diagnosis can flag depression or PTSD well above chance. This launched a surge of “distant supervision” studies that infer labels from hashtags, subreddit membership, or key phrases. Successive papers quickly pushed in domain F1 scores past 0.80, driven primarily by the advent of deep contextual language models such as BERT and RoBERTa.

Yet, subsequent work has revealed that these impressive numbers often mask underlying fragility. Harrigan, Aguirre, and Dredze (2020) demonstrated that models trained exclusively on Reddit data frequently collapse when evaluated on Twitter, noting that subtle platform artifacts like slang, emoji habits, and posting cadence can falsely drive accuracy metrics. Large scale reviews underscore the severity of the problem: Chancellor and De Choudhury (2020) found that 87% of 75 surveyed studies relied on purely heuristic labels, while Aldkheel and Zhou (2024) highlighted pervasive issues with small sample sizes, youth dominated cohorts, and a near absence of subgroup performance reports. Currently, rigorous privacy safeguards, open replication code, and clinically validated ground truth remain the exception rather than the rule.

Against this backdrop, the field is beginning to shift toward personalized, multi label, and interpretable models. Architectures such as PIE (Wu et al., 2024) attempt to integrate user specific linguistic baselines into transformers, yielding steady performance gains without prohibitive compute costs. This is an encouraging sign that tailoring model inputs can mitigate over fitting to population level quirks. The theoretical basis for our approach is rooted in “Digital Phenotyping” (Torous & Firth, 2018), which justifies the use of behavioral features as clinical proxies. By translating clinical symptoms into digital signals, such as interpreting affective instability as sentiment volatility, or sleep disruption as circadian deviation, we ground our computational model in established psychiatric indicators rather than arbitrary platform metrics. Addressing these gaps through rigorous cross platform validation, user centered explainability, and the tracking of biological proxies defines the frontier for the next generation of social media mental health research.

### 3. Methodology: The PAMHeD System

#### 3.1 Data Pipeline & Acquisition

The primary dataset was constructed by scraping Twitter/X user timelines utilizing the Nitter backend, targeting users with self disclosed diagnoses of Depression, Anxiety, or Bipolar disorder, alongside a matched cohort of Control users. To ensure data quality, we implemented strict language filtration using the `langdetect` library, supplemented with heuristic fall-

| Class Label    | Users        | Tweets         |
|----------------|--------------|----------------|
| Control (0)    | 1,460        | 175,240        |
| Depression (1) | 655          | 78,620         |
| Anxiety (2)    | 645          | 77,508         |
| Bipolar (3)    | 600          | 72,110         |
| <b>Total</b>   | <b>3,360</b> | <b>403,478</b> |

**Table 1:** Dataset statistics following language filtration and sanitization.

backs, dropping any user whose timeline exceeded a 30% non English threshold. Furthermore, a final infinity sanitization pass was applied to the dataset to prevent exploding gradients during Z score normalization. Recognizing the highly visual nature of modern social media, we implemented multimodal processing via BLIP (Bootstrapping Language Image Pre training) to generate descriptive captions for image attachments before appending them to their tweet text, thereby converting visual media into analyzable textual signals.

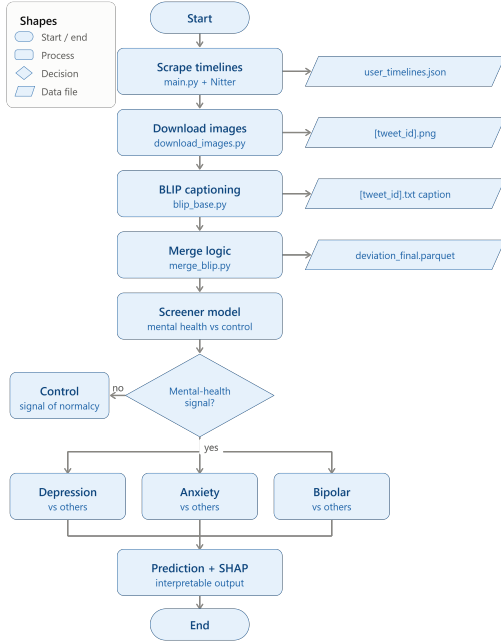
Table 1 details the composition of the final sanitized dataset utilized for model training and in domain evaluation.

#### 3.2 Feature Engineering: The “Delta” Vector

To operationalize the concept of personalization, we engineered a “Delta” vector for each tweet. This was calculated by establishing a behavioral baseline using the first 40% of a user’s timeline and then tracking subsequent deviations from that norm. These deviations were standardized into global Z scores to ensure comparability across users. The engineered features act as digital proxies for established biological and psychological markers. To capture Circadian Disruption (the “Insomnia” Signal) we calculated the circadian deviation (the z score of the posting hour versus the baseline median) alongside flags for late night posting. To measure Affective Instability (the “Volatility” Signal) we calculated the standard deviation of VADER sentiment scores over a 10 tweet rolling window, as well as tracking the frequency of abrupt polarity shifts. Finally, to quantify Cognitive and Social Withdrawal (the “Isolation” Signal) we measured the self focus ratio (the proportion of I/me/my pronoun usage), isolation lexicon hit scores, and multi scale drops in overall tweeting activity.

#### 3.3 Model Architectures & The “Delta” Advantage

We utilized `mental-roberta-base` as our text only Baseline classifier. While this pre trained model is highly capable of extracting latent stylistic and emotional patterns, such as a fatalistic tone, cognitive slowing, or absolutist phrasing, even in the absence of explicit mental health keywords, these linguistic signals in isolation are often weak and ambiguous. For ex-



**Figure 1:** System architecture overview showing the data flow from raw tweets through multimodal processing to final classification.

ample, a negative political tweet may simply indicate momentary frustration rather than clinical depression, causing the text only model to hit a distinct performance ceiling when faced with noisy, real world data.

To overcome this limitation, we designed the **AdvancedDeltaModel**. This custom architecture fuses the dense semantic vector (the [CLS] token embedding) extracted by RoBERTa with the dense behavioral feature vector via a learned Gated Fusion mechanism. The core design philosophy of this architecture is to combine multiple weak signals into a single high confidence conclusion. By combining a weakly negative textual signal with strong behavioral anomalies, such as erratic off hours posting with a high standard deviation, abnormally high posting volume, and highly volatile sentiment, the model acts as a multi modal detective. The gated mechanism is crucial, as it scales the behavioral input and prevents the numerical dominance of extreme behavioral values, allowing the network to confidently differentiate between benign textual complaints and genuine pathological deviations.

## 4. Experimental Method

### 4.1 Environment & Configuration

The experiments were conducted in an isolated Python 3.12 virtual environment on a local machine equipped with an NVIDIA RTX 3060 GPU to leverage CUDA acceleration. The environment was configured using standard data science and deep learning libraries: `torch` (v2.x), `transformers` (Hugging Face), `scikit-learn`, and `pandas`.

Model architecture configurations, including hyper-parameters, were held constant to ensure fair comparison between the Baseline and Delta variants: 5 training epochs, a learning rate of  $3 \times 10^{-5}$  with a 10% linear warmup schedule, and a batch size of 16. To prevent shortcut learning observed in early testing, robust regularization was applied using a dropout rate of 0.4, weight decay set to 0.01, and full 32 bit precision (FP16=False) to ensure stable gradient flow during the combination of text and numerical features.

### 4.2 Data Acquisition & Preparation

Two primary datasets were utilized:

1. **In Domain (Twitter/X):** We utilized a custom scraper (via the `ntscraper` library) to construct a dataset of approximately 3,000 users, split into self disclosed pathologies (Depression, Anxiety, Bipolar) and matched Controls based on activity rates. We mandated a minimum of 50 tweets per user. Raw image attachments were captioned using the Salesforce BLIP model. A preprocessing script (`merge_blip.py`) filtered out non English users, calculated individual baselines (first 40% of a user’s timeline), and appended computed Z score deviation features for the remaining tweets, exporting the data as a Parquet file.
2. **Out of Domain (Reddit):** To test cross platform generalizability, we acquired the publicly available eRisk 2018 Task 1 dataset, containing long form Reddit posts labeled for Depression vs. Control. A compilation script parsed the raw XML files, after which the exact same feature engineering script was applied to reconstruct the required behavioral feature vectors.

### 4.3 Execution & High Level Instructions

The experiment is designed as a sequential pipeline. The codebase and reproduction instructions are available in a public repository <https://github.com/Victorg514/PAMHED>

1. **Data Preprocessing:** Execute `python merge_blip.py` to ingest raw JSON timelines, compute features, perform a stratified user level train/val/test split (handling class imbalance via random undersampling of the majority class), and output the training ready Parquet dataset.
2. **Model Training (One vs Rest):** The experimental approach utilizes parameterized scripts to train binary classifiers. To reproduce the “Depression Expert” experiment, execute:

- `python train_binary_baseline.py -positive_class depression`
- `python train_binary_delta.py -positive_class depression`

This process is repeated for the “Screener” model using `-positive_class mental_health`. A `WeightedRandomSampler` was explicitly configured within the `PyTorch DataLoader` to guarantee balanced 50/50 batches during training.

3. **Evaluation:** Execute `python evaluate_binary_models.py -positive_class depression` to load the cached weights and generate classification reports on the held out test split. To replicate the cross platform experiment, execute `python evaluate_on_erisk.py`.
4. **Explainability Audit:** A Jupyter Notebook (`SHAP_demo.ipynb`) is provided to load a trained model, pass a sample instance, and generate SHAP waterfall plots to visualize feature contributions.

## 5. Experimental Results

The primary objective of these experiments was to observe the performance differences between a text only classifier and a personalized, multi modal classifier across varying class structures and platforms.

### 5.1 Experiment 1: Multi Class Observation (4 Class)

Table 2 displays the classification report generated when both models were tasked with differentiating all four classes simultaneously (Control, Depression, Anxiety, Bipolar).

**How to read this result:** The Delta model shows a substantial increase in Recall for the Control class (0.37 to 0.59), leading to a higher F1 score for that specific class. However, F1 scores for the three pathology classes decreased marginally in the Delta model compared to the Baseline. This conflicts with the experiment statement predicting uniform improvement across all classes.

### 5.2 Experiment 2: Binary Strategy (In Domain Twitter)

To address the low pathology scores observed in Experiment 1, the problem was restructured into One vs Rest (OvR) binary tasks. Table 3 reports the performance of models trained specifically as a “Screener” (Control vs. All Mental Health) and a “Depression Expert” (Depression vs. Others).

**How to read this result:** In both binary configurations, the Delta model yields higher F1 scores than the Baseline. Notably, in the Depression Expert model, the Delta variant demonstrates a significant increase in Recall (identifying 81% of true depression cases compared to 66% for the baseline), aligning with the experiment statement that personalization improves signal detection.

### 5.3 Experiment 3: Feature Explainability (SHAP)

**How to read this result:** To understand the reasoning driving the personalized “Screener” model, we conducted a global explainability audit using SHAP across a sample of high confidence predictions. The results, visualized in Figure 2, yield several critical insights into the model’s behavior.

First, an analysis of the textual stream reveals persistent “shortcut learning.” The model assigns the highest magnitude of importance to non semantic `<pad>` tokens, indicating that it is utilizing tweet length (inferred through the volume of padding required to reach `max_length`) as a primary diagnostic heuristic, rather than relying on deep semantic understanding.

Second, the behavioral stream corroborates this finding while also demonstrating the value of linguistic profiling. The most impactful feature overall is `len_dev` (deviation from the user’s baseline tweet length). This confirms that structural shifts in communication style are highly predictive. Importantly, the second most impactful feature is `self_focus` (the ratio of first person pronouns). This validates established psychological literature (e.g., Pennebaker, 2011) suggesting that an inward cognitive shift is a robust, quantifiable marker of psychological distress, successfully captured by the Delta model.

### 5.4 Experiment 4: Cross Platform Evaluation (Reddit eRisk)

Table 4 details the performance when the “Depression Expert” models, trained exclusively on Twitter data, were evaluated against an artificially balanced subset (50/50 Control/Depression) of the eRisk 2018 Reddit dataset.

**How to read this result:** The Baseline (text only) model achieves a higher overall F1 score and higher Recall on the out of domain data. The Delta model’s Recall drops significantly (to 17%), but it achieves the highest Precision (67%) in the experiment. This presents a mixed result regarding the experiment statement: the personalized model is more accurate when it makes a positive prediction, but it misses a vast majority of cases across platforms.

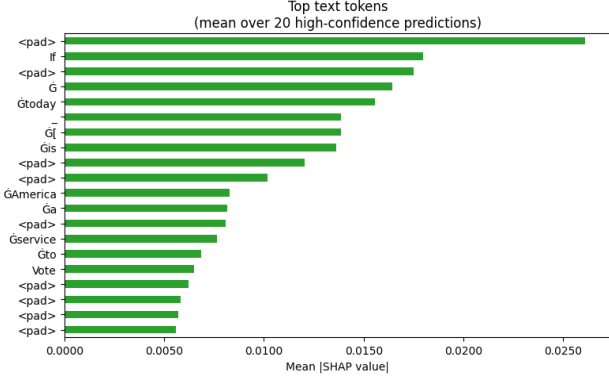
## 6. Conclusions

### 6.1 Interpretation of Results

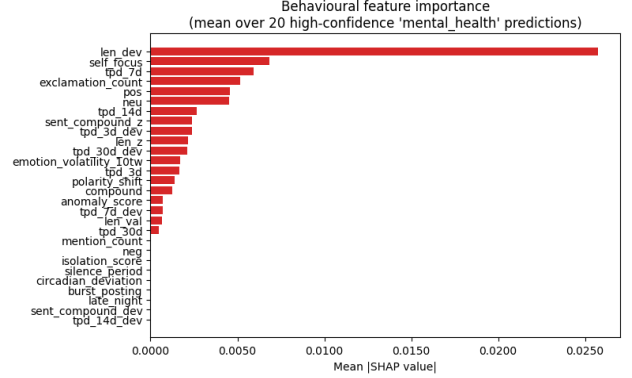
The experimental results of the PAMHeD system demonstrate that the effectiveness of personalized behavioral modeling in computational psychiatry is highly dependent on task formulation and platform consistency. The initial hypothesis posited that a personalized model would uniformly outperform a static text only approach. However, the data revealed a more

| Class            | Baseline (Text Only) |               |               | Delta (Personalized) |               |               |
|------------------|----------------------|---------------|---------------|----------------------|---------------|---------------|
|                  | Precision            | Recall        | F1 Score      | Precision            | Recall        | F1 Score      |
| Control          | 0.6322               | 0.3770        | 0.4723        | 0.5626               | 0.5940        | 0.5779        |
| Depression       | 0.2158               | 0.2490        | 0.2312        | 0.2187               | 0.2095        | 0.2140        |
| Anxiety          | 0.2295               | 0.3199        | 0.2672        | 0.2458               | 0.2341        | 0.2398        |
| Bipolar          | 0.2111               | 0.3054        | 0.2497        | 0.2267               | 0.2156        | 0.2210        |
| <b>Macro Avg</b> | <b>0.3222</b>        | <b>0.3128</b> | <b>0.3051</b> | <b>0.3134</b>        | <b>0.3133</b> | <b>0.3132</b> |

**Table 2:** In domain 4 class performance comparison.



(a) Textual semantic tokens.



(b) Behavioral (Delta) features.

**Figure 2:** Global SHAP summary plots showing mean absolute feature importance across high confidence predictions.

| Model    | Screener      |               | Dep. Expert   |               |
|----------|---------------|---------------|---------------|---------------|
|          | F1            | Rec. (MH)     | F1            | Rec. (Dep)    |
| Baseline | 0.6738        | 0.6783        | 0.3142        | 0.6671        |
| Delta    | <b>0.6927</b> | <b>0.7229</b> | <b>0.3244</b> | <b>0.8104</b> |

**Table 3:** In domain performance for binary OvR configurations. Screener is Control vs. all Mental Health; Depression Expert is Depression vs. Others.

| Model (Reddit Test)  | F1            | Prec.         | Rec.          |
|----------------------|---------------|---------------|---------------|
| Baseline (Text Only) | <b>0.4151</b> | 0.6130        | <b>0.3138</b> |
| Delta (Personalized) | 0.2768        | <b>0.6752</b> | 0.1741        |

**Table 4:** Cross platform performance evaluating Twitter trained models on balanced Reddit data.

nuanced reality regarding how digital phenotypes operate in practice.

In the multi class setting (Experiment 1), the personalized Delta model struggled to differentiate between specific mental health conditions. The behavioral signals indicating a “deviation from normal” such as erratic sleep patterns or sudden bursts of activity are symptomatic overlaps common to Anxiety, Depression, and Bipolar disorder. While the model could easily identify the stability of a Control user, it experienced severe inter class confusion among the pathologies. This indicates that while personalization is an excellent general anomaly detector, behavioral inten-

sity alone lacks the semantic specificity required for differential diagnosis.

Conversely, when the problem was reframed as targeted binary classification tasks (Experiment 2), the Delta model proved highly superior. The inclusion of noisy behavioral features acted as a powerful regularizer, successfully preventing the model collapse that frequently plagues highly imbalanced, text only setups. The “Screener” model (F1 0.69) proved particularly adept at establishing “normalcy” and triggering alerts upon deviation. This confirms that knowing a user’s baseline is incredibly valuable for initial triage and distress flagging.

Perhaps the most significant finding emerged from the cross platform experiment (Experiment 4), which revealed a fundamental precision recall trade off. The text only Baseline generalized far better broadly (yielding higher recall), proving that the semantic expression of depression, such as expressions of worthlessness or fatigue, is a relatively universal human constant across both Twitter and Reddit. Conversely, the Delta model’s recall collapsed out of domain. This strongly suggests that mechanical behavioral anomalies, such as daily posting frequency or specific interaction rates, are heavily platform dependent and do not transfer securely across different social media ecosystems. However, the Delta model retained an exceptionally high precision (67%) on the data it did flag. This implies that when the personalized model recognizes universal biological proxies, such as severe circadian disruption (insomnia) or intense affective insta-

bility over time, it acts as an exceptionally accurate, albeit highly conservative, diagnostic filter.

## 6.2 What Went Wrong & Lessons Learned

A critical lesson from this study was the underestimation of the multi class problem. Considerable computational resources and time were spent attempting to tune a 4 class model facing severe biological signal overlap. A simpler binary approach should have been adopted much earlier in the pipeline to isolate specific predictive features.

Furthermore, early iterations of our training pipeline suffered from severe “shortcut learning,” where models assigned artificially high importance to spurious artifacts like padding tokens rather than genuine semantic meaning. This vulnerability was only discovered via SHAP explainability audits. This highlights a profound danger in computational mental health research: trusting raw accuracy scores in black box models can lead to the deployment of systems that learn dataset mechanics rather than clinical realities. Integrating explainability tools must be a mandatory step in the validation process, not just a post hoc visualization.

## 6.3 Future Work

Based on the insights gleaned from these experiments, the development of personalized mental health detection systems must move beyond simple feature concatenation and address the nuances of signal overlap and platform dependence. Future exploration should focus on three primary avenues to enhance robustness and clinical validity.

First, immediate efforts must focus on refining the OvR Ensemble. While the individual binary “Expert” models demonstrated significant improvements in handling class imbalance and preventing model collapse, they currently exist as isolated classifiers. The next logical step is to complete the training and aggregation of all four expert binary models (Depression, Anxiety, Bipolar, and the Control Screener) to generate a final, hierarchical prediction system. By prioritizing the high confidence Screener model to first filter out stable users, and subsequently routing flagged individuals to the specialized pathology models, we hypothesize a substantial improvement in overall diagnostic accuracy. This complete ensemble approach must then be evaluated directly against the original, flawed multi class baseline to definitively quantify the performance lift gained through task isolation.

Second, the data pipeline must evolve to incorporate explicit Interaction Features. Currently, the model relies on the neural network to find correlations between independent behavioral metrics like posting frequency and sentiment scores. However, psychiatric symptoms are rarely isolated. By engineering features that explicitly combine these metrics prior to model ingestion, we can provide stronger, more clinically rele-

vant signals. For example, multiplying a High Activity score by an Extreme Positive Sentiment score could create a specific, calculated “Mania Indicator.” This explicit feature engineering would give the Bipolar Expert model a much cleaner signal to learn from, potentially resolving the inter class confusion where erratic behavior was previously indistinguishable from anxious insomnia.

Finally, the concept of the baseline itself requires further investigation through Cluster Based Baselines. The current PAMHeD system relies on strict per user normalization, calculating a unique baseline for every individual. Although highly personalized, this approach is vulnerable to individual noise and requires a substantial volume of historical data per user to be statistically reliable. Future work should experiment with grouping users into cohorts based on intrinsic traits, such as clustering users by their baseline writing style using Linguistic Inquiry and Word Count (LIWC) categories. By establishing a “group norm” for users with similar baseline emotional expression or formality, the model could calculate deviations against a more statistically stable cluster average. Determining whether group level normalization provides a more robust signal than isolated individual tracking is a critical next step for deploying these models at scale.

## 7. Ethics & Privacy Statement

The use of social media data for inferring mental health status raises significant ethical, privacy, and consent considerations. This study utilizes public Twitter/X data scraped via the Nitter interface and the publicly available eRisk 2018 dataset. No interaction with users occurred, and no interventions were staged. The target labels (Depression, Anxiety, Bipolar) were inferred strictly from public self disclosure statements.

To protect user anonymity, all datasets were de identified prior to model training. Unique user handles were stripped, and internal database identifiers were utilized exclusively. The objective of this research is strictly computational and methodological; it is not intended to develop a clinical diagnostic tool or to profile individuals.

We adhere to the ethical guidelines established by the computational psychiatry community, ensuring that predictive modeling does not infringe upon user privacy or autonomy.

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